

Zhuoran (Lucy) Sun

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EDUCATION

Carnegie Mellon University

Master of Computational Data Science (Analytics Track)

GPA: **4.04/4.0**; Courses: Deep Reinforcement Learning, Machine Learning, Cloud Computing, AI Engineering

Pittsburgh, PA

Expected: Dec 2025

Emory University

Bachelor of Business Administration, Information Systems, Double Major: Math & CS

GPA: **3.95/4.0** (High Distinction, ISOM Distinction Award, Beta Gamma Sigma)

Atlanta, GA

May 2021

WORK EXPERIENCE

Keyfactor (Cybersecurity Startup)

Cleveland, OH/Atlanta, GA

Data Engineer Intern

June 2025 – Aug 2025

- Led a cross-functional project to develop a Minimum Viable Product for a Post-Quantum Cryptography awareness-scoring system for Fortune 500 companies. The system integrates web scraping to collect timely public news for the companies and performs both traditional NLP and prompting Claude for sentiment analysis to enhance GTM strategy.
- Designed and led the implementation of data pipelines for Revenue Operations, automating data ingestion from Salesforce and ETL processes, which saved on average 5 hours of manual effort per product-level ARR report.
- Engineered and deployed an end-to-end application on Google Cloud Platform, collecting and cleaning geospatial data for thousands of US and EMEA leads. The system includes data collection, processing, and an interactive dashboard to enable geo-targeted sales analytics.

Truist Securities

Quantitative Research Analyst/Associate

Atlanta, GA

July 2021 – June 2024

Data Engineering

- Designed data models and led automation of daily PnL and risk reporting pipeline across fixed-income desk, reducing generation time from hours to minutes with Pandas, NumPy, Snowflake and Tableau, and reporting directly to C-level managers. Developed processes to perform quarterly data scrutiny checks across multiple databases on 50K+ records.
- Took the initiative to perform SQL query optimization on the annual trading volume and balance sheet analysis for all trading businesses and reduced query generation time from 1 hour to 15 minutes for each of 6 trading desks.

Tool Building

- Optimized RUNZ, an event-driven bond price listener using SpringBoot to capture Bloomberg quote messages, enabling price change alerts and pre and post trading time-series analysis for 200+ high-yield bonds concurrently.
- Built and distributed a Python API package on internal Artifactory to communicate between reporting and derivative pricing systems, ensuring stable ZMQ connectivity and secure compliance.

Research and Analytics

- Delivered a POC analysis for a quartile-based bond trading strategy, utilizing statistical and time-series analysis, including ARIMA, Regression, on 3M+ request for quote data over 3+ years as part of the E-trading strategy initiative.

PROJECTS

Twitter Recommendation with Cloud Solutions, CMU

Pittsburgh, PA (Jan. 2025 – May 2025)

- Engineered a Spark ETL pipeline in Scala on 1TB+ Twitter data in Azure Databricks, designing an optimized data schema to calculate user recommendation scores and improve downstream micro-service performance
- Deployed production-level micro-services (Vert.x; Java; gRPC; REST) in Kubernetes clusters with Terraform, Docker, and Helm charts on AWS ECS, AWS EKS. Compared backend between AWS RDS MySQL, DynamoDB, and ElastiCache Redis
- Fine-tuned the clusters to achieve latency within 5ms over a 3-hour live test and ranked top 5 in the leaderboard

Model Pruning for Efficient Deep Learning, CMU

Pittsburgh, PA (Mar. 2025 – May 2025)

- Performed pruning on a large CNN, achieving >94% sparsity while losing <1% accuracy, optimizing efficiency-accuracy tradeoffs for large-scale ML models, combining activation-based and network slimming methods

E-commerce Database Analysis Research, Emory University

Atlanta, GA (Aug. 2019 – July 2020)

- Analyzed 40M+ cart events from a major e-commerce platform using time-series models, random forests, LSTM, and DBSCAN in SciPy and PyTorch, boosting database efficiency and achieving 91% ROC in checkout prediction
- Paper published: OLTP In Real Life: A Large-scale Study of Database Behavior in Modern Online Retail at MASCOTS 2021

Sales System, The Emory Wheel

Atlanta, GA (Sept. 2017 – June 2020)

- Migrated sales data from Google Sheets to a Django-based online sales system with PostgreSQL database deployed on Heroku, enabling report generation, analytics, visualization, and commission calculations, saving the sales manager 50% of time on analytics and organization